

**VIT**Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

REG.NO.:

SCHOOL OF SOCIAL SCIENCES AND LANGUAGES
CONTINUOUS ASSESSMENT TEST - II
FALL SEMESTER 2025-2026

SLOT:A1+TA1+TAA1

Programme Name & Branch : B.Tech.,
Course Code and Course Name : BMAT201L and Complex Variables and Linear Algebra
Faculty Name(s) : Common question paper
Class Number(s) : Common question paper
Date of Examination : 05.10.2025
Exam Duration : 90 minutes **Maximum Marks: 50**

General instruction(s):

- Answer All Questions
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)
- Course Outcomes (Type the CO statements covered in this question paper. Use the CO number as per the syllabus copy)

CO3: Evaluate real integrals using techniques of contour integration

CO4: Use the power of inner product and norm for analysis

CO5: Use matrices and transformations for solving engineering problems

Q. No	Question	M	CO	BL
1.	a) If $f(z) = \frac{1-e^z}{1+e^z}$, then find the location of singularities and also discuss the nature of singularity when $z \rightarrow \infty$. b) Using Cauchy Integral formula, evaluate $\int_C \frac{3z^3}{(z-1)(z^2+9)} dz$, where C is the circle given by $ z-2 =2$.	3+7	CO3	2
2.	Using contour integration, evaluate $\int_{-\infty}^{\infty} \frac{\sin x}{x^2+x+1} dx$.	10	CO3	3
3.	Using Cayley Hamilton theorem, find the inverse of the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$. Also find the eigen values of A , A^T and A^{-1} .	10	CO5	3
4.	Find the solution of the following system of equations given by $x + 2y - 3z + 4w = 12$ $2x + 2y - 2z + 3w = 10$ $y + z = -1$ $x - y + z - 2w = -4$ using Gauss Jordan elimination method.	10	CO5	3
5.	a) Check whether $w = \{(x, y, z) / \frac{y}{x} = \sqrt{2}\}$ is a subspace of R^3 . b) Check whether the set of vectors given by $\{(-1, 0, -1), (3, 1, 6), (8, 3, 17)\}$ form a basis of R^3 or not.	5+5	CO4	2

consider

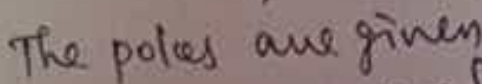
and function not defined as $z \rightarrow \infty$. (1M)
 $\therefore z \rightarrow \infty$ is a removable singularity.

↳ a circle with $C(2,0), r=2$.

$$\int_C \frac{f(z)}{z-a} dz = 2\pi i \cdot f(a), \text{ where}$$

$$\therefore 2\pi i \cdot f(a)$$

$$= 2\pi i \cdot \frac{3}{105} = \left(\frac{3}{5} \pi i \right)$$



by $z = \frac{-1 \pm i\sqrt{3}}{2}$

Which only $z = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$ lies inside C .

From fig.

$$(1M) \int_C f(z) e^{iz} dz = \int_{-R}^R f(x) \cdot e^{ix} dx + \int_{\gamma_0} f(z) e^{iz} dz$$

$$2\bar{\lambda}_1 \leq \text{Res}$$

(imp residue theorem)

Allowing $R \rightarrow \infty$,

$$\int_{-\infty}^{\infty} \frac{e^{ix}}{x^2 + x + 1} dx = 2\pi i \cdot \text{Res}\left(-\frac{1}{2} + \frac{i}{\sqrt{3}}\right)$$
(11) L (1)

reason: consider $|f(z)| = \left| \frac{1}{z^2 + 2z + 1} \right| \rightarrow 0$ as $|z| \rightarrow \infty$

(1M) $\therefore \int_{C_R} \frac{1}{z^2 + 2z + 1} \cdot e^{iz} dz \rightarrow 0$ as $R \rightarrow \infty$.

Res $f(-\frac{1}{2} + \frac{i\sqrt{3}}{2}) = \lim_{z \rightarrow -\frac{1}{2} + \frac{i\sqrt{3}}{2}} \frac{1 \cdot e^{iz} \cdot (z + \frac{1}{2} - \frac{i\sqrt{3}}{2})}{(z + \frac{1}{2} + \frac{i\sqrt{3}}{2})(z + \frac{1}{2} - \frac{i\sqrt{3}}{2})}$

(2M) $= \frac{e^{i(-\frac{1}{2} + \frac{i\sqrt{3}}{2})}}{i\sqrt{3}} = \frac{1}{i\sqrt{3}} \cdot e^{-i/2 - \frac{\sqrt{3}}{2}}$

$\therefore \textcircled{1} \Rightarrow \int_{-\infty}^{\infty} \frac{1}{x^2 + x + 1} \cdot e^{ix} dx = 2\pi i \cdot \frac{1}{i\sqrt{3}} \cdot e^{-i/2 - \frac{\sqrt{3}}{2}}$ (1M)

$\int_{-\infty}^{\infty} \frac{1}{x^2 + x + 1} [\cos x + i \sin x] dx = \frac{2\pi}{\sqrt{3}} \cdot e^{-\frac{\sqrt{3}}{2}} \cdot \left\{ \cos \frac{1}{2} - i \sin \frac{1}{2} \right\}$ (1M)

Comparing Imaginary parts on both sides,

$\int_{-\infty}^{\infty} \frac{\sin x}{x^2 + x + 1} \cdot dx = -\frac{2\pi}{\sqrt{3}} e^{-\frac{\sqrt{3}}{2}} \sin \frac{1}{2}$ (1M)

③ The char. equation of A is given by $|A - \lambda I| = 0$

(1M) $\lambda^3 + \lambda^2 - 5\lambda - 5 = 0$ — (1) Using C-H theorem

(3M) $A^3 + A^2 - 5A - 5I = 0$ $A^2 = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (1M)

$\Rightarrow A^{-1} = \frac{1}{5} [A^2 + A - 5I]$

(3M) $= \frac{1}{5} \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -5 \end{bmatrix}$

The eig(A) = $-1, -\sqrt{5}, \sqrt{5} = \text{eig}(A^{-1})$ (3M)

$\text{eig}(A^{-1}) = -1, -\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}$

4) In matrix form

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 2 & 2 & -2 & 3 \\ 0 & 1 & 1 & 0 \\ 1 & -1 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 12 \\ 10 \\ -1 \\ -4 \end{bmatrix} \quad (1M)$$

consider

$$A/b = \left[\begin{array}{cccc|c} 1 & 2 & -3 & 4 & 12 \\ 2 & 2 & -2 & 3 & 10 \\ 0 & 1 & 1 & 0 & -1 \\ 1 & -1 & 1 & -2 & -4 \end{array} \right]$$

$$Ax = b$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_4 \rightarrow R_4 - R_1$$

$$= \left[\begin{array}{cccc|c} 1 & 2 & -3 & 4 & 12 \\ 0 & -2 & 4 & -5 & -14 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & -3 & 4 & -6 & -16 \end{array} \right]$$

$$R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -2 & 4 & -5 \\ 0 & 1 & 1 & 0 \\ 0 & -3 & 4 & -6 \end{bmatrix}$$

$$\begin{cases} x = 1 \\ y = 0 \\ z = -1 \\ w = 2 \end{cases}$$

$$= \left[\begin{array}{cccc|c} 1 & 2 & -3 & 4 & 12 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & -2 & 4 & -5 & -14 \\ 0 & -3 & 4 & -6 & -16 \end{array} \right]$$

$$R_1 \rightarrow R_1 - 2R_2$$

$$R_3 \rightarrow R_3 + 2R_2$$

$$R_4 \rightarrow R_4 + 3R_2$$

$$= \left[\begin{array}{cccc|c} 1 & 0 & -5 & 4 & 14 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 6 & -5 & -16 \\ 0 & 0 & 7 & -6 & -19 \end{array} \right]$$

$$=$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -5 & 4 & 14 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 1 & -5/6 & -8/3 \\ 0 & 0 & 7 & -6 & -19 \end{array} \right]$$

$$R_1 \rightarrow R_1 + 5R_3$$

$$R_2 \rightarrow R_2 - R_3$$

$$R_4 \rightarrow R_4 - 7R_3$$

$$\approx \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1/6 & 2/3 \\ 0 & 1 & 0 & 5/6 & 5/3 \\ 0 & 0 & 1 & -5/6 & -8/3 \\ 0 & 0 & 0 & -1/6 & -1/3 \end{array} \right]$$

$$\approx \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1/6 & 2/3 \\ 0 & 1 & 0 & 5/6 & 5/3 \\ 0 & 0 & 1 & -5/6 & -8/3 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right]$$

$$\approx \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right]$$

$$5) a) \left. \begin{aligned} u &= (x_1, y_1, z_1) \\ v &= (x_2, y_2, z_2) \end{aligned} \right\} \in W. \Rightarrow \frac{y_1}{x_1} = \sqrt{2}; \frac{y_2}{x_2} = \sqrt{2}.$$

T.P. $u+v \in W$
 $u+v = (x_1+x_2, y_1+y_2, z_1+z_2)$

To prove $\frac{y_1+y_2}{x_1+x_2} = \sqrt{2}.$

(2M) LHS (i.e.) $\frac{\sqrt{2}x_1 + \sqrt{2}x_2}{x_1+x_2} = \sqrt{2}. \Rightarrow \underline{u+v \in W}$

For any scalar α , $\alpha u = \alpha(x_1, y_1, z_1)$
 $= (\alpha x_1, \alpha y_1, \alpha z_1)$

To prove $\frac{\alpha y_1}{\alpha x_1} = \sqrt{2}.$

(2M) LHS $\frac{\alpha y_1}{\alpha x_1} = \frac{y_1}{x_1} = \sqrt{2}. \Rightarrow \underline{\alpha u \in W}$ (1M)

Q. W is a subspace of $\mathbb{R}^3$.

5) b) $\alpha(-1, 0, -1) + \beta(3, 1, 6) + \gamma(8, 3, 17) = 0$. (1M)

$$\left. \begin{aligned} ① \leftarrow -\alpha + 3\beta + 8\gamma &= 0 \\ ② \leftarrow 0\alpha + \beta + 3\gamma &= 0 \\ ③ \leftarrow -\alpha + 6\beta + 17\gamma &= 0 \end{aligned} \right\} \text{ from ① \& ②, } \frac{\alpha}{9-8} = \frac{\beta}{-(-3)} = \frac{\gamma}{-1} = k$$

$\Rightarrow \alpha = k; \beta = 3k; \gamma = -k.$

Using α, β, γ in ③,

$$\begin{aligned} -k + 18k - 17k &= 0 \\ \Rightarrow 18k - 18k &= 0 \\ 0 &= 0. \end{aligned}$$

Put $k=1, \Rightarrow \alpha=1; \beta=3, \gamma=-1$

$\Rightarrow$ The given vectors are not L.I.

$\therefore$ The given set of vectors does not form a basis of $\mathbb{R}^3$. (1M)